Pharmaniaga Atenolol Tablet

DESCRIPTION
Pharmaniaga Atenolol Tablet 50 mg
An orange, round, film-coated tablet with a breakline on one side and plain on the other.
Pharmaniaga Atenolol Tablet 100 mg
An orange, round, biconvex, film-coated tablet with Pharmaniaga icon on one side and breakline on the other.

Pharmaniaga Atenolol Tablet 50 mg/ 100 mg should not be split into half.

COMPOSITION
Pharmaniaga Atenolol Tablet 50 mg
Each film-coated tablet contains Atenolol 50 mg.
Pharmaniaga Atenolol Tablet 100 mg
Each film-coated tablet contains Atenolol 100 mg.

PHARMACODYNAMICS
Pharmacotherapeutic group: Beta-blocking agents, plain, selective, ATC code: CO7A B03

Atenolol is a beta-blocker which is beta1-selective, (i.e. acts preferentially on beta1-adrenergic receptors in the heart). Selectivity decreases with increasing dose.

Atenolol is without intrinsic sympathomimetic and membrane-stabilising activities and as with other beta blockers, has negative inotropic effects (and is therefore contraindicated in uncontrolled heart failure).

As with other beta-blockers, the mode of action of atenolol in the treatment of hypertension is unclear. It is probably the action of atenolol in reducing cardiac rate and contractility which makes it effective in eliminating or reducing the symptoms of patients with angina.

It is unlikely that any additional ancillary properties possessed by S (- atenolol, in comparison with the racemic mixture, will give rise to different therapeutic effects. Atenolol is effective and well-tolerated in most ethnic populations although the response may be less in black patients.

Atenolol is effective for at least 24 hours after a single oral dose. The drug facilitates compliance by its acceptability to patients and simplicity of dosing. Atenolol is compatible with diuretics, other hypotensive agents and antianginals. ice it acts preferentially on beta-receptors in the heart, atenolol may, with care, be used successfully in the treatment of patients with respiratory disease, who cannot tolerate non-selective beta-blockers.

Atenolol is an additional treatment to standard coronary care.

PHARMACOKINETICS

Absorption
Absorption of atenolol following oral dosing consistent but incomplete (approximately 40-50%) with peak plasma concentrations occurring 2-4 hours after dosing. Atenolol blood levels are consistent and subject to little variability. There is no significant hepatic metabolism of atenolol and more than 90% of that absorbed reaches the systemic circulation unaltered. Food decreases bioavailability of atenolol by 20%. Atenolol displays linear pharmacokinetics profile.

Distribution
Atenolol penetrates tissues poorly due to its low lipid solubility and its concentration in brain tissue is low. Plasma protein binding is low (approximately 3%).

Elimination
The plasma half-life is about 6 hours but this may rise in severe renal impairment since the kidney is the major route of elimination.

INDICATION
For the treatment of hypertension, angina pectoris, acute myocardial infarction and cardiac arrhythmias.

CONTRAINDICATIONS
This medicine, as with other beta-blockers, should not be used in patients with any of the following:
- hypersensitivity to the active substance, or to any of the excipients
- cardiogenic shock
- uncontrolled heart failure
- sick sinus syndrome
* second-or third-degree heart block
- untreated phaeochromocytoma
- metabolic acidosis
- bradycardia (<45 bpm)
- hypotension
- severe peripheral arterial circulatory disturbances

ADVERSE REACTIONS
Blood and lymphatic system disorders
Rare: Purpura, thrombocytopenia

Psychiatric disorders
Uncommon: Sleep disturbances of the type noted with other beta-blockers
Rare: Mood changes, nightmares, confusion, psychoses and hallucinations

Nervous system disorders
Rare: Dizziness, headache, paraesthesia

Eye disorders
Rare: Dry eyes, visual disturbances

Cardiac disorders
Common: Bradycardia
Rare: Heart failure deterioration, precipitation of heart block

Vascular disorders
Common: Cold extremities
Rare: Postural hypotension which may be associated with syncope, intermittent claudication may be increased if already present, in susceptible patients Raynaud's phenomenon

Respiratory, thoracic and mediastinal disorders
Rare: Bronchospasm may occur in patients with bronchial asthma or a history of asthmatic complaints, pulmonary embolism

Gastrointestinal disorders
Common: Gastrointestinal disturbances
Rare: Dry mouth

Hepatobiliary disorders
Uncommon: Elevations of transaminase levels
Rare: Hepatic toxicity including intrahepatic cholestasis

Skin and subcutaneous tissue disorders
Rare: Alopecia, psoriasiform skin reactions, exacerbation of psoriasis, skin rashes
Not Known: Hypersensitivity reactions, including angioedema and urticaria

Musculoskeletal and connective tissue disorders
Not known: Lupus-like syndrome

Reproductive system and breast disorders
Rare: Impotence

General disorders and administration site conditions
Common: Fatigue

Investigations
Very rare: An increase in ANA (Antinuclear Antibodies) has been observed, however the clinical relevance of this is not clear

Endocrine metabolic
Thyrotoxicosis

Discontinuance of the drug should be considered if, according to clinical judgement, the well-being of the patient is adversely affected by any of the above reactions.

WARNING AND PRECAUTION
This medicine, as with other beta-blockers:
- Should not be withdrawn abruptly. The dosage should be withdrawn gradually over a period of 7-14 days, to facilitate a reduction in beta-blocker dosage. Patients should be followed during withdrawal, especially those with ischaemic heart disease.

- When a patient is scheduled for surgery, and a decision is made to discontinue beta-blocker therapy, this should be done at least 24 hours prior to the procedure. The risk-benefit assessment of stopping beta-blockade should be made for each patient.

If treatment is continued, an anaesthetic with little negative inotropic activity should be selected to minimise the risk of myocardial depression. The patient may be protected against vagal reactions by intravenous administration of atropine.

- Although contraindicated in uncontrolled heart failure may be used in patients whose signs of heart failure have been controlled. Caution must be exercised in patients whose cardiac reserve is poor.

- May increase the number and duration of angina attacks in patients with Prinzmetal's angina due to unopposed alpha-receptor mediated coronary artery vasoconstriction. Atenolol is a beta1-selective beta-blocker; consequently, its use maybe considered although utmost caution must be exercised.

- Although contraindicated in severe peripheral arterial circulatory disturbances may also aggravate less severe peripheral arterial circulatory disturbances.

- Due to its negative effect on conduction time, caution must be exercised if it is given to patients with first-degree heart block.

- May mask the symptoms of hypoglycaemia, in particular, tachycardia.

- May mask the signs of thyrotoxicosis (e.g. tachycardia). Abrupt withdrawal may precipitate thyroid storm.

- Will reduce heart rate as a result of its pharmacological action. In the rare instances when a treated patient develops symptoms which may be attributable to a slow heart rate and the pulse rate drops to less than 50-55 bpm at rest, the dose should be reduced.

• May cause a more severe reaction to a variety of allergens when given to patients with a history of anaphylactic reaction to such allergens. Such patients may be unresponsive to the usual doses of adrenaline (epinephrine) used to treat the allergic reactions.

• May cause a hypersensitivity reaction including angioedema and urticaria.

• Should be used with caution in the elderly, starting with a lesser dose.

• Since atenolol is excreted via the kidneys, dosage should be reduced in patients with a creatinine clearance of below 35ml/min/1.73 m^2.

• Although cardioselective (beta1) beta-blockers may have less effect on lung function than non-selective beta-blockers, as with all beta-blockers, these should be avoided in patients with reversible obstructive airways disease, unless there are compelling clinical reasons for their use. Where such reasons exist, atenolol may be used with caution. Occasionally some increase in airways resistance may occur in asthmatic patients however, and this may usually be reversed by commonly used dosage of bronchodilators such as salbutamol or isoprenaline.

• Withdrawal of concomitant clonidine therapy; atenolol may increase risk of rebound hypertension; atenolol should be discontinued several days before clonidine is withdrawn.

• As with other beta-blockers, in patients with a phaeochromocytoma, an alpha-blocker should be given concomitantly.

PREGNANCY AND LACTATION

Caution should be exercised when atenolol is administered during pregnancy or to a woman who is breast-feeding.

Pregnancy

Atenolol crosses the placental barrier and appears in the cord blood. There is no data on the use of atenolol in the first trimester and the possibility of foetal injury cannot be excluded. Atenolol has been used under close supervision for the treatment of hypertension in the third trimester. Administration of atenolol to pregnant women in the management of mild to moderate hypertension has been associated with intra-uterine growth retardation.

The use of ate lol in women who are, or may become, pregnant requires that the anticipated benefit be weighed against the possible risks, particularly in the first and second trimesters, since beta-blockers, in general, have been associated with a decrease in placental perfusion which may result in intra-uterine deaths, immature and premature deliveries.

Breast-feeding

There is significant accumulation of atenolol in breast milk. Atenolol diffuses into breast milk in concentrations similar to or higher than those in maternal blood. Neonates born to mothers who are receiving atenolol at parturition or breast-feeding may be at risk of hypoglycaemia and bradycardia.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

Atenolol has no or negligible influence on the ability to drive and use machines. However, it should be taken into account that occasionally dizziness or fatigue may occur.

DRUG INTERACTIONS

• Combined use of beta-blockers and calcium channel blockers with negative inotropic effects, e.g. verapamil and diltiazem, can lead to an exaggeration of these effects particularly in patients with impaired ventricular function and/or sinoatrial or atrioventricular conduction abnormalities. This may result in severe hypotension, bradycardia and cardiac failure. Neither the beta-blocker nor the calcium channel blocker should be administered intravenously within 48 hours of discontinuing the other.

• Concomitant therapy with dihydropyridines, e.g. nifedipine, may increase the risk of hypotension, and cardiac failure may occur i patients with latent cardiac insufficiency.

• Digitalis glycosides, in association with beta-blockers, may increase atrioventricular conduction time.

• Concurrent use of atenolol and clonidine may result in increased risk of sinus bradycardia; exaggerated clonidine withdrawal response (acute hypertension).

• Beta-blockers may exacerbate the rebound hypertension which can follow the withdrawal of clonidine. If the two drugs are co-administered, the beta-blocker should be withdrawn several days before discontinuing clonidine. If replacing clonidine by beta-blocker therapy, the introduction of beta-blockers should be delayed for several days after clonidine administration has stopped.

• Class I anti-arrhythmic drugs (e.g. disopyramide) and amiodarone may have a potentiating effect on atrial-conduction time and induce negative inotropic effect.

• Concomitant use of sympathomimetic agents, e.g. adrenaline (epinephrine), may counteract the effect of beta-blockers.

• Concomitant use with insulin and oral antidiabetic drugs (e.g. acarbose, gliclazide, metformin) may lead to the intensification of the blood sugar lowering effects of these drugs. Symptoms of hypoglycaemia, particularly tachycardia, may be masked.

• Concomitant use of prostaglandin synthetase-inhibiting drugs, e.g. ibuprofen and indometacin, may decrease the hypotensive effects of beta-blockers.

• Caution must be exercised when using anaesthetic agents with atenolol. The anaesthetist should be informed and the choice of anaesthetic should be an agent with as little negative inotropic activity as possible. Use of beta-blockers with anaesthetic drugs may result in attenuation of the reflex tachycardia and increase the risk of hypotension. Anaesthetic agents causing myocardial depression are best avoided.

• Concurrent use of atenolol and alpha-1 adrenergic blockers (e.g. alfuzosin, doxazosin, prazosin and terazosin) may result in an exaggerated hypotensive response to the first dose of the alpha blocker.

• Concurrent use of atenolol and quinidine may result in bradycardia, hypotension.

• Concurrent use of atenolol and St John's Wort may result in decreased effectiveness of beta-adrenergic blockers.

• Concurrent use of atenolol and beta-2 agonists (e.g. fenoterol, formoterol and salmeterol) may result in severe bronchospasm and decreased effectiveness of the beta-2 agonist.

• Concurrent use of atenolol and antacids (e.g. aluminium, calcium or magnesium containing products) may result in reduced effectiveness of atenolol.

• Concurrent use of atenolol and warfarin may result in risk of increased prothrombin time or INR.

• Concurrent use of atenolol and chlorpromazine may result in hypotension and/or phenothiazine toxicity.

• Concurrent use of atenolol and methyldopa may result in exaggerated hypertensive response, tachycardia, or arrhythmias during physiologic stress or exposure to exogenous catecholamines.

• Concurrent use of atenolol and glycopyrrolate may result in increased plasma concentrations of atenolol.

DOSAGE AND ADMINISTRATION

The dose must always be adjusted to individual requirements of the patients, with the lowest possible starting dosage.

Usual adult dose:

Angina Pectoris:

Most patients with angina pectoris will respond to 100 mg given orally once or 50 mg given twice daily. It is unlikely that additional benefit will be gained by increasing the dose.

Hypertension:

One tablet daily. Most patients respond to 100 mg daily given orally as a single dose. Some patients, however, will respond to 50 mg given as a single daily dose. The effect will be fully established after one to two weeks. A further reduction in blood pressure may be achieved by combining atenolol with other antihypertensive agents. For example, co-administration of atenolol with a diuretic, provides a highly effective and convenient antihypertensive therapy.

Cardiac arrhythmias

Initially controlled intravenously. Having controlled the arrhythmias with intravenous atenolol, a suitable oral maintenance dosage is 50-100 mg daily, given as a single dose.

Acute Myocardial Infarction:

In suitable patients, initially controlled intravenously, followed by atenolol 50 mg orally about 15 minutes later, provided no untoward effects have occurred from the intravenous dose. This should be followed by a further 50 mg orally 12 hours after the intravenous dose and then 12 hours later by 100 mg orally, once daily. If bradycardia and/or hypotension requiring treatment, or any other untoward effects occur, atenolol should be discontinued.

Late intervention after acute myocardial infarction:

For patients who present some days after suffering an acute myocardial infarction an oral dose of atenolol (100 mq daily) is recommended for long-term prophylaxis of myocardial infarction.

Elderly patients:
Dosage requirements may be reduced, especially in patients with impaired renal function.

Renal failure
Since atenolol is excreted via the kidneys the dosage should be reduced in cases of severe impairment of renal function.

No significant accumulation of atenolol occurs in patients who have a creatinine clearance greater than 35 ml/min/1.73m^2 (normal range is 100-150 ml/min/1.73m^2).

For patients with a creatinine clearance of 15-35 ml/min/1.73m^2 (equivalent to serum creatinine of 300-600 micromol/litre) the oral dose should be 50 mg daily.

For patients with a creatinine clearance of <15 ml/min/1.73m^2 (equivalent to serum creatinine of >600 micromol/litre) the oral dose should be 25 mg daily or 50 mg on alternate days.

Patients on haemodialysis should be given 50 mg orally after each dialysis; this should be done under hospital supervision as marked falls in blood pressure can occur.

Usual paediatric dose:
There is no paediatric experience with atenolol and for this reason it is not recommended for use in children.

OVERDOSAGE
Symptoms:
The symptoms of overdosage may include bradycardia, hypotension, acute cardiac insufficiency and bronchospasm.

Treatment:
General treatment should include: close supervision, treatment in an intensive care ward, the use of gastric lavage, activated charcoal and a laxative to prevent absorption of any drug still present in the gastrointestinal tract, the use of plasma or plasma substitutes to treat hypotension and shock. The use of haemodialysis or haemoperfusion may be considered.

Excessive bradycardia can be countered with atropine 1-2 mg intravenouslv and/or a cardiac pacemaker. If necessary, this may be followed by a bolus dose of glucagon 10 mg intravenously. If required, this may be repeated or followed by an intravenous infusion of glucagon 1-10 mg/hour depending on response. If no response to glucagon occurs or if glucagon is unavailable, a beta-adrenoceptor stimulant such as dobutamine 2.5 to 10 micrograms/kg/minute by intravenous infusion may be given. Dobutamine, because of its positive inotropic effect could also be used to treat hypotension and acute cardiac insufficiency. It is likely that these doses would be inadequate to reverse the cardiac effects of beta-blocker blockade if a large overdose has been taken. The dose of dobutamine should therefore be increased if necessary to achieve the required response according to the clinical condition of the patient.

Bronchospasm can usually be reversed by bronchodilators.

ROUTE OF ADMINISTRATION
Oral.

STORAGE CONDITIONS
Store below 30°C.
Protect from light.

SHELF LIFE
Product should not be used beyond the expiry date imprinted on the product packaging.

PRESENTATION
Pharmaniaga Atenolol Tablet 50 mg
In blisters of 100 and 250 tablets.

Pharmaniaga. Atenolol Tablet 100 mg
In blisters of 30, 100, 250 and 500 tablets.

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PRODUCT REGISTRATION HOLDER/ MANUFACTURER:
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